

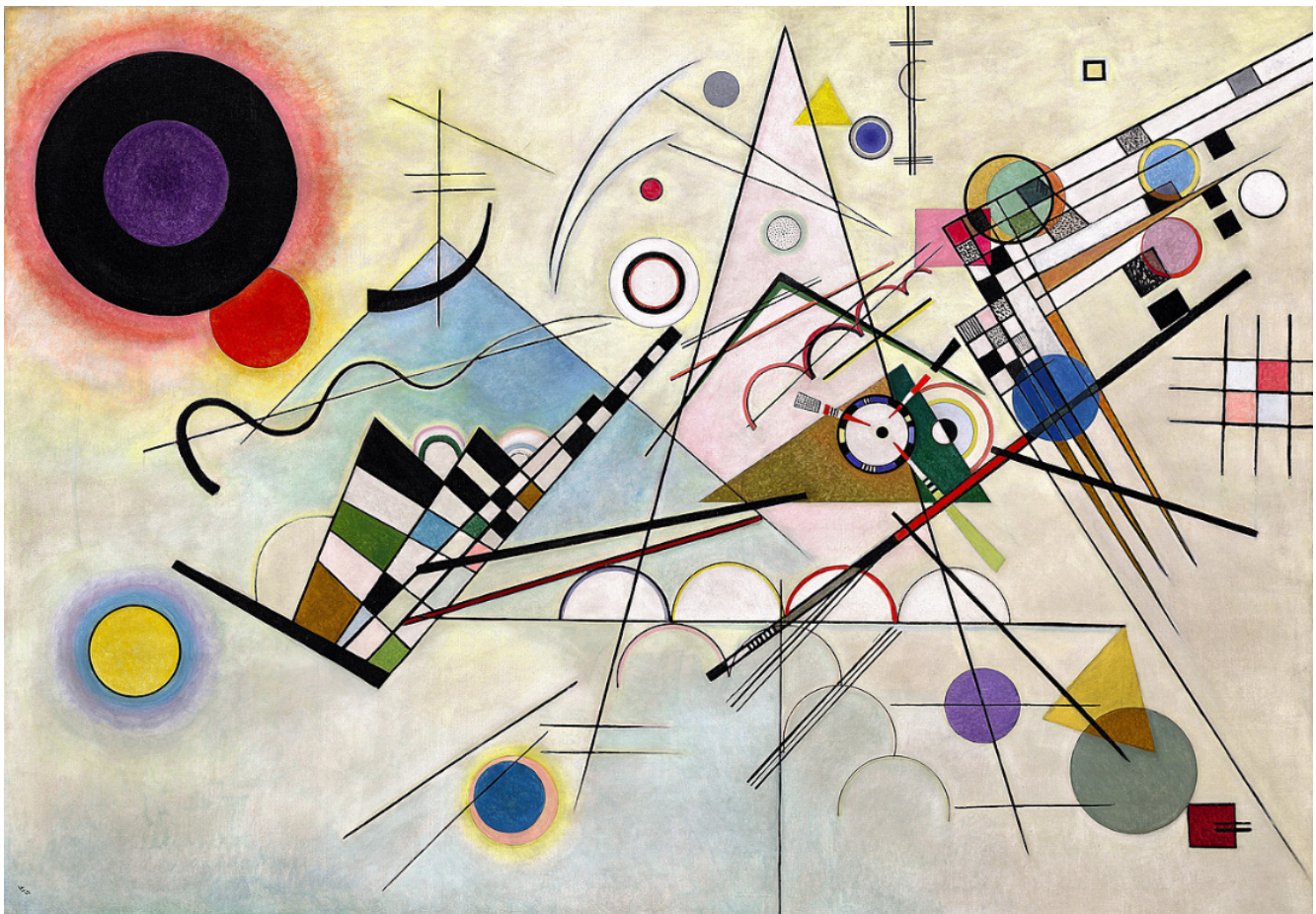
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How Do We Measure an Image?

LMTH 2040 — Calculus: A Multidisciplinary Approach

August 26, 2026

Eugene Lang College of Liberal Arts



Key Idea

What proportion of this image is occupied by **circular forms**?

There is no formula provided.

You may draw on the image, measure it, divide it into pieces, count things, approximate, or use a calculator.

Work with your group. Be prepared to explain your method.

1. Decide what you are measuring

Before calculating anything, decide what you mean by a **circular form**.

What counts? What does not count? How will you treat overlapping, partially hidden, or outlined forms?

2. Invent a method

Describe a procedure for estimating the proportion of the image occupied by circular forms.

Your method should be clear enough that another group could reproduce it.

3. Make an estimate

Our estimate is

or approximately

%

of the image.

4. What could go wrong?

Identify at least two sources of uncertainty, ambiguity, or error.

1.

2.

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Refining the Measurement

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5. Make the method better

Suppose we place a square grid over the image.

For each square, we decide whether it belongs to the region we are measuring. This gives an estimate of the proportion of the whole image occupied by that region.

What happens if we repeat the procedure using increasingly fine grids?

$$10 \times 10, \quad 100 \times 100, \quad 1000 \times 1000, \quad \dots$$

Let the corresponding estimates be

$$A_{10}, \quad A_{100}, \quad A_{1000}, \quad \dots$$

Problem

What do you expect this sequence of estimates to do as the grid becomes finer?

Problem

Could this procedure eventually produce an *exact* answer?
Why or why not?

6. A different question

A computer can calculate

$$A_{10}, \quad A_{100}, \quad A_{1000}, \quad A_{10000}, \quad \dots$$

but it cannot literally complete infinitely many measurements.

Key Idea

Can an infinite mathematical process produce an exact finite answer?

Before you leave

Respond briefly to each question.

MEASURE What had to be decided before your group could produce a number?

APPROXIMATE Why might increasingly fine measurements approach something more precise?

QUESTION Can mathematics describe something that neither you nor a computer could ever literally finish computing?

Image revealed after discussion: Wassily Kandinsky, *Composition VIII*, 1923. Oil on canvas, 140×201 cm.